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Inventor(s)	Laurin; Robert S. et al.

Single height adjustment and control assembly for walk-behind outdoor power equipment

Abstract

Outdoor power equipment includes a frame, a power head, a transmission, a pair of rear drive wheels, and a wheel height adjustment mechanism. The power head is coupled to the frame. The transmission is coupled to the frame and operatively coupled to the power head. The transmission includes a drive rotatable member configured to rotate about a first axis. The pair of rear drive wheels are coupled at respective ends of a rear drive axle so that the rear drive axle, and the pair of rear drive wheels coupled thereto, rotate about a second axis different than the first axis. The wheel height adjustment mechanism includes a first bell crank pivotally coupled to the frame about the first axis. A first end of the first bell crank is coupled to the rear drive axle.

Inventors: Laurin; Robert S. (Durhamville, NY), Lynch; E. Mark (Munnsville, NY), Camenga; Derrick R. (Sherburne, NY)

Applicant: Briggs & Stratton, LLC (Wauwatosa, WI)

Family ID: 1000008821480

Assignee: Briggs & Stratton, LLC (Wauwatosa, WI)

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Primary Examiner: Torres; Alicia

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/549,078, filed Aug. 23, 2019, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/725,642, filed on Aug. 31, 2018 and U.S. Provisional Patent Application No. 62/745,515, filed on Oct. 15, 2018, all of which are incorporated herein by reference in their entireties.

BACKGROUND

(1) The present disclosure relates generally to outdoor power equipment such as lawnmowers, and, more particularly, to walk-behind lawnmowers having a single height adjustment arrangement for simultaneous height adjustment for all four wheels of the lawnmower, as well as a user control assembly for controlling the speed and direction of the lawnmower.

(2) The use of outdoor power equipment such as walk-behind lawnmowers is prevalent among both commercial and residential operators due to, e.g., the mowers' relative light weight and maneuverability, convenience when mowing small lawns and/or difficult-to-reach locations, compact storage and transport requirements, etc. Many variations of walk-behind lawnmowers exist, from small, single-blade, electrically-driven mowers generally utilized in residential settings, to larger, multi-blade, engine-driven mowers used most often by commercial operators.

(3) Depending on grass heights, mowing conditions, user preferences, etc., it is generally desirable for the height-of-cut settings of a lawnmower to be adjustable. For most ride-on lawnmowers and zero turn radius lawnmowers, a suspended mower deck is provided, allowing for the height-of-cut settings to be adjusted simply by raising or lowering the mower deck relative to the frame of the lawnmower. However, for most walk-behind lawnmowers, height-of-cut settings are adjusted by individually raising or lowering the height of each of the four wheels of the lawnmower. While such a height adjustment method allows the entire mower deck, blade(s), and power head (e.g., an engine) to raise or lower together, thereby avoiding the complexities of the belt-and-pulley systems utilized on most ride-on and zero turn radius lawnmowers, the need to adjust the height at each wheel is both time consuming and prone to user error. If the height of each wheel is not set equally,

the blade(s) will become angled relative to the cutting surface, leading to an uneven cut.

(4) This patent document described devices that are intended to address the issues discussed above and/or other issues.

SUMMARY

(5) One embodiment relates to outdoor power equipment including a frame, a power head, a transmission, a pair of rear drive wheels, and a wheel height adjustment mechanism. The power head is coupled to the frame. The transmission is coupled to the frame and operatively coupled to the power head. The transmission includes a drive rotatable member configured to rotate about a first axis. The pair of rear drive wheels are coupled at respective ends of a rear drive axle so that the rear drive axle, and the pair of rear drive wheels coupled thereto, rotate about a second axis different than the first axis. The wheel height adjustment mechanism includes a first bell crank pivotally coupled to the frame about the first axis. A first end of the first bell crank is coupled to the rear drive axle.

(6) Another embodiment relates to a walk-behind lawnmower having a frame, a power head, a mower deck, a transmission, a front wheel, a pair of rear drive wheels, and a wheel height adjustment mechanism. The power head is coupled to the frame. The mower deck is coupled to the frame. The transmission is coupled to the frame and is operatively coupled to the power head. The transmission includes a drive member configured to rotate about a first axis. The pair of rear drive wheels are coupled at respective ends of a rear drive axle. The rear drive axle is coupled to the drive member so that the rear drive axle, and the pair of rear drive wheels coupled thereto, rotate about a second axis. The wheel height adjustment mechanism includes at least one rear bell crank pivotally coupled to the frame about the first axis. A first end of the at least one rear bell crank is coupled to the rear drive axle such that movement of the rear drive axle relative to the frame does not affect tension of a drive between the drive member and the rear drive axle.

(7) In accordance with an aspect of the disclosure, an outdoor power equipment unit is disclosed. The outdoor power equipment unit includes a frame and a power head coupled to the frame. The unit also includes a hydrostatic transmission coupled to the frame and operatively coupled to the power head, wherein the hydrostatic transmission comprises a drive gear configured to rotate about a first axis. A pair of rear drive wheels may also be coupled at respective ends of a rear drive axle, wherein the rear drive axle is operably coupled to a driven gear, and further wherein the pair of rear drive wheels, the rear drive axle, and the driven gear are configured to rotate about a second axis different than the first axis. A drive chain may be operably coupled to both the drive gear and the driven gear. Also, a wheel height adjustment mechanism may be provided, wherein the wheel height adjustment mechanism comprises a first bell crank pivotally coupled to the frame about the first axis, and further wherein a first end of the first bell crank is coupled to the rear drive axle.

(8) According to another aspect of the disclosure, a walk-behind lawnmower is disclosed. The walk-behind lawnmower may include a frame and a power head coupled to the frame. The lawnmower may also include a mower deck coupled to the frame. A hydrostatic transmission may be coupled to the frame and operatively coupled to the power head, wherein the hydrostatic transmission comprises a drive gear configured to rotate about a first axis. A pair of front wheels may also be provided. A pair of rear drive wheels may be coupled at respective ends of a rear drive axle, wherein the rear drive axle is operably coupled to a driven gear, and further wherein the pair of rear wheels, the rear drive axle, and the driven gear are configured to rotate about a second axis. A drive chain operably may be coupled to both the drive gear and the driven gear. A wheel height adjustment mechanism may also be provided, wherein the wheel height adjustment mechanism comprises at least one rear bell crank pivotally coupled to the frame about the first axis, and further wherein a first end of the at least rear one bell crank is coupled to the rear drive axle such that movement of the rear drive axle relative to the frame does not affect tension of the drive chain between the drive gear and the driven gear.

(9) According to another aspect of the disclosure, a control assembly for use with outdoor power

equipment is disclosed. The control assembly may include a control housing. The control assembly may also include a speed control lever extending from the control housing, wherein the speed control lever is coupled to an asymmetric body portion. The asymmetric body portion may be pivotally coupled to the control housing, and the asymmetric body may have a plurality of projections formed on a radial surface the asymmetric body portion. The control assembly may also include a forward speed control handle. The forward speed control handle may be coupled to one or more primary arcuate brackets. The one or more primary arcuate brackets may be pivotally coupled to the control housing. Additionally, the one or more primary arcuate brackets may be limited in rotational travel by at least one of the plurality of projections formed on the radial surface of the asymmetric body portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of a walk-behind lawnmower having a single height adjustment configuration in accordance with an aspect of the disclosure;
- (2) FIG. 2 is a partial rear perspective view of the walk-behind lawnmower illustrated in FIG. 1;
- (3) FIG. 3 is a partial front perspective view of the walk-behind lawnmower illustrated in FIG. 1;
- (4) FIG. 4 is a partial side view of the walk-behind lawnmower illustrated in FIG. 1;
- (5) FIG. 5 is a partial side view of components of a height adjustment configuration and wheel drive configuration of the walk-behind lawnmower illustrated in FIG. 1;
- (6) FIG. 6 is a partial bottom view of components of a height adjustment configuration and wheel drive configuration of the walk-behind lawnmower illustrated in FIG. 1;
- (7) FIG. 7 is a partial top view of components of a height adjustment configuration, wheel drive configuration, and mower blade drive configuration of the walk-behind lawnmower illustrated in FIG. 1;
- (8) FIG. 8 is a partial front perspective view of a walk-behind lawnmower having a single height adjustment configuration in accordance with another aspect of the disclosure;
- (9) FIG. 9 is a partial front perspective view of a user control assembly for a walk-behind lawnmower in accordance with an aspect of the disclosure;
- (10) FIG. 10 is another partial front perspective view of the user control assembly of FIG. 9;
- (11) FIG. 11 is a side cross-sectional view of the user control assembly of FIG. 9;
- (12) FIG. 12 is an underside view of the user control assembly of FIG. 9;
- (13) FIG. 13 is another partial front perspective view of the user control assembly of FIG. 9;
- (14) FIG. 14 is another partial front perspective view of the user control assembly of FIG. 9;
- (15) FIG. 15 is a transmission linkage assembly in accordance with an aspect of the disclosure; and
- (16) FIG. 16 is a side view of the transmission linkage assembly of FIG. 15.

DETAILED DESCRIPTION

- (17) The following description is made for the purpose of illustrating the general principles of the present system and method and is not meant to limit the inventive concepts claimed in this document. Further, particular features described in this document can be used in combination with other described features in each of the various possible combinations and permutations.
- (18) Unless otherwise specifically defined in this document, all terms are to be given their broadest possible interpretation including meanings implied from the specification as well as meanings understood by those skilled in the art and/or as defined in dictionaries, treatises, etc.
- (19) It must also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless otherwise specified. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. All publications mentioned in this document are

incorporated by reference. Nothing in this document is to be construed as an admission that the embodiments described in this document are not entitled to antedate such disclosure by virtue of prior invention. As used herein, the term “comprising” means “including, but not limited to”. Additionally, use the term “couple”, “coupled”, or “coupled to” may imply that two or more elements may be directly connected or may be indirectly coupled through one or more intervening elements.

(20) In this document, position-identifying terms such as “vertical”, “horizontal”, “front”, “rear”, “top”, and “bottom” are not intended to limit the invention to a particular direction or orientation, but instead are only intended to denote relative positions, or positions corresponding to directions shown when a mower is oriented as shown in the Figures.

(21) Referring to FIGS. 1-7, outdoor power equipment in the form of a walk-behind lawnmower **10** is shown. For the sake of clarity, some components typically associated with walk-behind lawnmowers have been omitted from FIGS. 1-7, including mower deck and drive system shielding, drive system linkages, etc. However, one of ordinary skill in the art will readily recognize that these components may be incorporated into walk-behind lawnmower **10** in accordance with the embodiments described in this document.

(22) As shown in FIGS. 1-2, walk-behind lawnmower **10** includes a pair of rear drive wheels **13a**, **13b**, as well as a pair of front wheels **14a**, **14b**. A power head in the form of an internal combustion engine **12** is provided to power both the rear drive wheels **13a**, **13b** and the one or more mower blades (not shown) carried by a mower deck **15**. Specifically, rear drive wheels **13a**, **13b** may be driven in both forward and reverse directions by a hydrostatic transmission **26** which, as will be described further herein, is operably coupled to engine **12** via a belt-and-pulley arrangement. The hydrostatic transmission **26** is configured to drive the rear drive wheels **13a**, **13b** by way of a drive chain **27**, as will also be described further below. The one or more mower blades within mower deck **15** are similarly driven by the engine **12** via another belt-and-pulley arrangement. However, it is to be understood that a power head other than an internal combustion engine may be utilized to provide power to the rear drive wheels **13a**, **13b** and/or the one or more mower blades. For example, one or more battery-powered electric motors may be utilized in lieu of the engine **12**.

(23) Referring still to FIGS. 1-2, walk-behind lawnmower **10** also includes user controls **16** provided on an end of a pair of extended arms **17a**, **17b**, which are coupled to respective sides of a frame **28**. While not shown or described in detail, it is to be understood that user controls **16** may include, e.g., forward and reverse directional drive and speed controls, stationary handle(s), a blade-engagement control, etc. Furthermore, while not shown, the directional drive and speed controls of the user controls **16** may be coupled to a hydrostatic transmission interface **30** (see FIG. 3) by way of one or more control linkages.

(24) As described above, the mower deck of most residential and light commercial walk-behind lawnmowers remains fixed relative to the lawnmower's frame or, in many cases, substantially forms the lawnmower's frame). Thus, in order to adjust the height-of-cut, conventional walk-behind lawnmowers necessitate an adjustment in wheel height at each individual wheel. Walk-behind lawnmower **10** is no different, as mower deck **15** is coupled to, and remains stationary relative to, frame **28**. However, as will be described in further detail below, unlike previous methods for height-of-cut adjustment for walk-behind lawnmowers, walk-behind lawnmower **10** includes a single height adjustment mechanism for the simultaneous and coordinated height adjustment of all four wheels **13a**, **13b**, **14a**, **14b** relative to the frame **28**.

(25) As shown in FIGS. 1-4, rear drive wheels **13a**, **13b** are operably coupled to a rear bell crank **18**, with rear bell crank **18** configured to be pivotally coupled to frame **28** about a pivot joint **29**. Specifically, a common rear drive axle **32** upon which rear drive wheels **13a**, **13b** are rotatably mounted is coupled to a first end portion of rear bell crank **18**, while a second end portion of rear bell crank **18** is coupled to a retractable pin member **19**. Retractable pin member **19** is configured to selectively engage one of a plurality of incrementally-spaced through-holes **20** formed in a portion

of the frame **28**. In some embodiments, retractable pin member **19** may be spring-biased so as to hold pin member **19** in a “closed” position (i.e., a position engaged with one of the plurality of through-holes **20**). Thus, when the user wishes to change the height of rear drive wheels **13a**, **13b** relative to frame **28**, the user may simply retract pin member **19** from one of through-holes **20** and pivot rear bell crank **18** about pivot joint **29** until pin member **19** is aligned with another of through-holes **20** corresponding to a desired height.

(26) While not shown in FIGS. **1-4**, the frame **28** may include indicia (e.g., height-of-cut in inches, incremental numerical indicators, etc.) at or near each of through-holes **20** so as to provide the user with a visual understanding of the possible height-of-cut adjustments. Due to the inherent nature of rear bell crank **18**, as the user pivots the rear bell crank **18** from one through-hole **20** to another, the rear drive wheels **13a**, **13b** move along an arcuate path such that raising the height-of-cut causes rear drive wheels **13a**, **13b** to move both downward and rearward relative to the user controls, while lowering the height-of-cut causes rear drive wheels **13a**, **13b** to move both upward and forward relative to the user controls. The user may then lock rear drive wheels **13a**, **13b** in a desired position by engaging the pin member **19** with a desired through-hole **20**.

(27) As is shown in FIG. **6**, a second bell crank **18** is pivotally coupled to frame **28** at a location near rear drive wheel **13a**, thereby allowing rear drive wheels **13a**, **13b** to be displaced equally when adjustments to height-of-cut are made. For this reason, the second bell crank **18** does not necessitate its own retractable pin member, allowing for a single height adjustment mechanism to be utilized. Furthermore, while the respective bell cranks **18** are coupled by the common rear drive axle **32**, each bell crank **18** may also be coupled by a rod **31**. Specifically, referring to FIG. **3**, FIG. **5**, and FIG. **6**, rod **31** is configured to extend from a mounting location **42** at an end of rear bell crank **18** to a similar mounting location at an end of the opposite bell crank **18**. In this way, rod **31** provides for additional structural support between the respective bell cranks. Furthermore, rod **31** is positioned in such a way so as to provide at least some protection to various components of the drive train coupled to rear drive axle **32**, which will be described in further detail below.

(28) In addition to the simultaneous height adjustment of both of rear drive wheels **13a**, **13b** relative to frame **28** via a single height adjustment mechanism, walk-behind lawnmower **10** also provides for the simultaneous height adjustment of both of the front wheels **14a**, **14b** via the same height adjustment mechanism. That is, referring to FIGS. **1-6**, a pair of height adjustment linkages **21a**, **21b** extend from the respective rear bell cranks (i.e., rear bell crank **18** and the corresponding opposite bell crank (shown in FIG. **6**)) to a pair of respective front bell cranks **23a**, **23b** which are pivotally coupled to respective front wheels **14a**, **14b**. For example, as is shown in FIGS. **3-5**, a first end of a linkage **21b** is pivotally coupled to rear bell crank **18** at a pivot point **39**, while a second, opposite end of linkage **21b** is pivotally coupled to a first end of a front bell crank **23b** at a pivot point **38b**. Front bell crank **23b** is itself configured to pivot about an axis **37** upon a bracket **24b**, with bracket **24b** extending substantially from a front portion of mower deck **15**. The front wheel **14b** is, in turn, configured to rotate about an axis **36b** located at a second, opposite end of front bell crank **23b**. While not described herein, it is to be understood that various components on the opposite side of walk-behind lawnmower **10** (i.e., front wheel **14a**, bell crank **23a**, axis **36a**, pivot point **38a**, bracket **24a**) operate in a same or similar manner. In this way, movement of rear bell crank **18** (and the other, opposite rear bell crank) imparts corresponding movement on front bell cranks **23a**, **23b** via the respective linkages **21a**, **21b**. This, in turn, provides for corresponding arcuate movement of front wheels **14a**, **14b** simultaneous to movement of rear drive wheels **13a**, **13b** during height-of-cut adjustments, all while using a single height adjustment mechanism provided by the interface between pin member **19** and through-holes **20**. Furthermore, with this arcuate movement, not only are the front wheels **14a**, **14b** and the rear drive wheels **13a**, **13b** simultaneously adjusted in height relative to the frame **28**, but the wheelbase between respective front wheels **14a**, **14b** and rear drive wheels **13a**, **13b** remains constant, regardless of the height-of-cut selected.

(29) Referring to FIGS. 3-4, respective front bell cranks **23a**, **23b** are shown as being coupled to respective brackets **24a**, **24b** along an inner side of respective front wheels **14a**, **14b**, while a straight pivot arm **25** is provided along an outer side of each of front wheels **14a**, **14b**. However, in other embodiments, it is to be understood that front bell cranks **23a**, **23b** may instead be provided along the outer side of each of front wheels **14a**, **14b**, or a pair of bell cranks may be provided along both the inner and outer sides of each of front wheels **14a**, **14b**. Additionally and/or alternatively, while a pair of linkages **21a**, **21b** is shown, it is to be understood that, in some embodiments, a single linkage may extend between one of the rear bell cranks **18** and one of the front bell cranks **23a**, **23b** if the front bell cranks **23a**, **23b** are otherwise coupled together (i.e., via a rod or other solid linkage extending directly between front bell cranks **23a**, **23b**).

(30) As is also shown in FIGS. 1-4, each linkage **21a**, **21b** may be coupled to a respective extension spring **22a**, **22b**, wherein each extension spring **22a**, **22b** is configured to assist in the raising and lowering of rear drive wheels **13a**, **13b** and front wheels **14a**, **14b** relative to frame **28**. Specifically, referring to FIG. 3, a first end of extension spring **22b** is coupled to linkage **21b** at a mounting bracket **34**, while a second, opposite end of extension spring **22b** is configured to be coupled to a threaded, adjustable receiver **35**, which is itself coupled to a mounting plate **33** extending from a top surface of mower deck **15**. While not shown, it is to be understood that extension spring **22a** is similarly configured. In this way, as the user removes pin member **19** to adjust the height-of-cut, the extension springs **22a**, **22b** act to aid the user in raising the height-of-cut (i.e., lowering the wheels relative to the frame **28**) by effectively pulling the linkages **21a**, **21b** and rear bell cranks **18** in a forward direction. Conversely, the extension springs **22a**, **22b** also may aid in the lowering of the height-of-cut by dampening the upward movement of the wheels, which, without the extension springs **22a**, **22b**, would otherwise be abrupt due to the overall weight of the frame **28**, the transmission **26**, the engine **12**, etc.

(31) Next, referring to FIGS. 5-6, more detailed views of various components of the drive train of walk-behind lawnmower **10** is shown. As described above, walk-behind lawnmower **10** is driven by a hydrostatic transmission **26**, which is itself powered by an internal combustion engine **12** via a drive pulley **50**. Similar to internal combustion engine **12**, hydrostatic transmission **26** is coupled to frame **28** in a fixed position. Depending upon the user inputs, the hydrostatic transmission **26** may drive the rear drive axle **32** in either a forward or reverse direction. The rear drive axle **32** has a driven gear **43** non-rotatably coupled thereon via a differential **60**, while the hydrostatic transmission **26** includes a drive gear **44** configured to rotate based on a user-selected speed and direction. A drive chain **27** provides a connection between the drive gear **44** and driven gear **43**, thereby imparting rotational motion on rear drive axle **32** (and, in turn, rear drive wheels **13a**, **13b**) based on user inputs to the hydrostatic transmission **26**. A tensioning sprocket **61** may also be provided, which may aid in maintaining proper tension on drive chain **27**, as well as aid in preventing drive chain **27** from dropping from drive gear **44** and/or driven gear **43**. Furthermore, while a drive chain **27** is shown, it is to be understood that an alternative driving member, such as, e.g., a belt, may be utilized. Similarly, while drive gear **44** and driven gear **43** are shown, it is to be understood that an alternative rotatable member, such as, e.g., a drive pulley and a driven pulley, may be utilized.

(32) With this drive train configuration, the rear drive wheels **13a**, **13b** may be raised and lowered via the respective rear bell cranks **18** relative to frame **28** without adversely affecting the tension of drive chain **27** between driven gear **43** and drive gear **44**. That is, as bell cranks **18** partially rotate about pivot joint **29** during a height-of-cut adjustment, the rear drive wheels **13a**, **13b** do not move in a strictly vertical fashion. Rather, rear drive wheels **13a**, **13b** are displaced in an arcuate fashion centered around axis A shown in FIG. 6. Referring still to FIG. 6, due to the substantially coaxial alignment of pivot joint **29** with drive gear **44** of hydrostatic transmission **26** about an axis A, and the substantially coaxial alignment of driven gear **43**, differential **60**, and rear drive axle **32** about an axis B, the arc through which the rear drive axle **32** and rear drive wheels **13a**, **13b** travel during

height-of-cut adjustments does not adversely affect the tension of drive chain **27**, thereby allowing the rear drive wheels **13a**, **13b** to move relative to the hydrostatic transmission **26** without altering the overall drive capabilities of the walk-behind lawnmower **10**.

(33) Next, referring to FIGS. **6-7**, in accordance with an aspect of the disclosure, belt-and-pulley arrangements for the drive of both the hydrostatic transmission **26** and the mower blades of the mower deck **15** are shown. As described above, the mower deck **15** is configured to include one or more mower blades (not shown), which are configured to cut grass and other vegetation, expelling the cuttings out of the mower deck **15** through a guard **49**. In the specific configuration shown in FIG. **7**, mower deck **15** is configured to have two mower blades (not shown), with each mower blade coupled to a respective driven pulley **56**. However, it is to be understood that more or fewer mower blades (and, hence, driven pulleys) may be utilized.

(34) Drive pulley **50**, as detailed above, is coupled to the engine **12**. As shown in FIG. **7**, a first driven pulleys **56** is directly coupled to the drive pulley **50** by way of a main belt **55**, while the other of the driven pulleys **56** is coupled to the first driven pulley **56** by way of a secondary belt **59**. A number of idler pulleys **53**, **57**, **58** are provided so as to maintain belt tension and optimal belt-pulley surface contact. Thus, as the drive pulley **50** rotates at high speeds when the user engages the mower blades via the user controls, the mower blades within mower deck **15** correspondingly rotate at high speeds to provide ideal cutting performance.

(35) In addition to the drive pulley **50** driving the various mower blades, drive pulley **50** also selectively drives the hydrostatic transmission **26** via a drive belt **52** and an intervening tensioning pulley **51**. Accordingly, similar to the drive of the mower blades, the user is able to selectively control the operation of the hydrostatic transmission **26** in order to impart forward or reverse motion on rear drive wheels **13a**, **13b**, thereby controlling the speed and/or direction of the walk-behind lawnmower **10**.

(36) As was noted above, the mower deck **15** is fixed relative to frame **28**, as are the engine **12** and the hydrostatic transmission **26**. Thus, even with changes in height-of-cut made by movement of rear drive wheels **13a**, **13b** and front wheels **14a**, **14b** upon their respective bell cranks, the angles of all belts and pulleys in the arrangement shown in FIG. **7** do not change. Thus, unlike suspended mower decks, which require relatively complex belt-and-pulley arrangements and spacing considerations to account for angular changes from the drive pulley to the driven pulleys based on deck height changes, the height-of-cut arrangement described herein with respect to FIGS. **1-7** does not necessitate such complex belt-and-pulley arrangements and spacing considerations, as the mower deck **15**, engine **12**, and hydrostatic transmission **26** are fixed relative to frame **28** and, thus, move in concert with one another during height-of-cut adjustments.

(37) While the walk-behind lawnmower **10** described above pertains to a light commercial-grade walk-behind lawnmower utilizing a hydrostatic transmission and belt-driven mower deck, the height-of-cut adjustment configurations described herein may also be applicable to other forms of walk-behind lawnmowers, including those utilizing a single mower blade driven directly by a power head (e.g., an internal combustion engine, an electric motor, etc.). Furthermore, while walk-behind lawnmower **10** illustrates a rear-wheel drive lawnmower, the concept is applicable to a front-wheel drive lawnmower or a push mower without assisted drive capabilities.

(38) Next, referring to FIG. **8**, a partial view of a walk-behind lawnmower **70** having a single height adjustment configuration in accordance with another aspect of the disclosure is shown. For the purposes of clarity, numerous components of the walk-behind lawnmower **70** have been omitted from the view shown in FIG. **8**. For example, it is to be understood that walk-behind lawnmower **70** may include a pair of rear drive wheels, an engine, a hydrostatic transmission, control handles, etc., similar to those found on walk-behind lawnmower **10** shown and described above with respect to FIGS. **1-7**.

(39) Walk-behind lawnmower **70** includes a frame **72**, upon which a power head and hydrostatic transmission (not shown) are fixedly coupled. Also fixedly coupled to frame **72** is a mower deck

74, which allows frame **72** and mower deck **74** (as well as the power head and hydrostatic transmission) to move in concert with one another during height-of-cut adjustments. However, as will be described in further detail below, unlike previous methods for height-of-cut adjustment for walk-behind lawnmowers, walk-behind lawnmower **70**, like walk-behind lawnmower **10**, includes a single height adjustment mechanism for the simultaneous and coordinated height adjustment of all four wheels relative to the frame **72**.

(40) As shown in FIG. **8**, and similar to walk-behind lawnmower **10** described above, the rear drive wheels (not shown) are configured to be operably coupled to a rear bell crank **84**, with rear bell crank **84** being pivotally coupled to frame **72** about a pivot joint **85**. The rear drive wheels are pivotally mounted upon a drive axle (not shown) about a pivot joint **86** at a first end portion of rear bell crank **84**, while a second end portion of rear bell crank **84** is coupled to a retractable pin member **87**. Retractable pin member **87** is configured to selectively engage one of a plurality of incrementally-spaced through-holes **88** formed in a portion of the frame **72**. In some embodiments, retractable pin member **87** may be spring-biased so as to hold pin member **87** in a “closed” position (i.e., a position engaged with one of the plurality of through-holes **88**). Thus, when the user wishes to change the height of rear drive wheels relative to frame **72**, the user may simply retract pin member **87** from one of through-holes **88** and pivot rear bell crank **84** about pivot joint **85** until pin member **87** is aligned with another of through-holes **88** corresponding to a desired height. Also, while not shown in FIG. **8**, it is to be understood that the drive axle is coupled to a second bell crank pivotally coupled to frame **72** at a location on the opposite side of frame **72** as rear bell crank **84**, thereby allowing rear drive wheels to be displaced equally when adjustments to height-of-cut are made. For this reason, the second bell crank (not shown) does not necessitate its own retractable pin member, allowing for a single height adjustment mechanism to be utilized.

(41) Due to the inherent nature of rear bell crank **84**, as the user pivots the rear bell crank **84** from one through-hole **88** to another, the rear drive wheels move along an arcuate path such that raising the height-of-cut causes rear drive wheels to move both downward and rearward relative to the user controls, while lowering the height-of-cut causes rear drive wheels to move both upward and forward relative to the frame **72** and to the user controls. The user may then lock rear drive wheels in a desired position by engaging the pin member **87** with a desired through-hole **88**.

(42) Referring still to FIG. **8**, in addition to the simultaneous height adjustment of both of rear drive wheels relative to frame **72** via a single height adjustment mechanism, walk-behind lawnmower **70** also provides for the simultaneous height adjustment of a pair of front caster wheel assemblies, which include front caster wheels **76a**, **76b**, via the same height adjustment mechanism. That is, a pair of height adjustment linkages **73a**, **73b** extend from the respective rear bell cranks (i.e., rear bell crank **84** and the corresponding opposite bell crank (not shown)) to a respective first leg of a pair of front bell cranks in the form of Y-shaped cranks **80a**, **80b** which are also pivotally coupled to respective wheel brackets **79a**, **79b** about respective second legs. Wheel brackets **79a**, **79b** are configured to rotatably support the caster wheels **76a**, **76b** via respective bearings **78a**, **78b** coupled to respective yokes **77a**, **77b**, each of which are configured to allow caster wheels **76a**, **76b** to rotate 360° about a vertical axis through bearings **78a**, **78b**.

(43) Each of the front Y-shaped cranks **80a**, **80b** is also pivotally coupled to a respective deck bracket **82a**, **82b**, which extends from a surface of the mower deck **74**. In this way, when the pair of rear bell cranks **84** are moved during desired height-of-cut adjustment, the pair of height adjustment linkages **73a**, **73b** impart corresponding movement on the front Y-shaped cranks **80a**, **80b** relative to mower deck **74**, with the Y-shape allowing for arcuate movement of the front caster wheels **76a**, **76b** in direct proportion to the arcuate movement of the rear drive wheels. Thus, proportional height adjustment of all four wheels of the walk-behind lawnmower **70** is attainable via a single height adjustment mechanism, while also maintaining a substantially constant wheel base between the front caster wheels **76a**, **76b** and the rear drive wheels (not shown), regardless of the height-of-cut selected.

(44) In addition to the front Y-shaped cranks **80a**, **80b**, the height adjustment configuration for the front caster wheels **76a**, **76b** further includes a respective pair of front pivoting linkages **81a**, **81b**, with front pivoting linkages **81a**, **81b** being pivotally coupled to wheel brackets **79a**, **79b** and deck brackets **82a**, **82b** at locations vertically in-line with the pivot locations of respective Y-shaped cranks **80a**, **80b**. In this way, Y-shaped cranks **80a**, **80b** and front pivoting linkages **81a**, **81b** act to substantially form a 4-bar linkage-type arrangement. With such a 4-bar linkage-type arrangement, the vertical alignment of front caster wheels **76a**, **76b** remains substantially constant through the vertical axis of bearings **78a**, **78b**, regardless of the selected height of front caster wheels **76a**, **76b** and the rear drive wheels. Thus, with no angular cant to the front caster wheels **76a**, **76b** relative to the frame **72** and mower deck **74** during height-of-cut adjustments, the front caster wheels **76a**, **76b** will not be prone to “wobble” dependent upon the selected height-of-cut, thereby improving contact with the ground and alignment of the front caster wheels **76a**, **76b** relative to the desired direction of travel.

(45) As is also shown in FIG. 8, at least one of the height adjustment linkages **73a**, **73b** may be coupled to an extension spring **75**, wherein the extension spring(s) **75** is configured to assist in the raising and lowering of rear drive wheels and front caster wheels **76a**, **76b** relative to frame **72**. In this way, as the user removes pin member **87** to adjust the height-of-cut, the extension spring(s) **75** may act to aid the user in raising the height-of-cut (i.e., lowering the wheels relative to the frame **72**) by effectively pulling on at least one of the height adjustment linkages **73a**, **73b** and rear bell cranks **84** in a forward direction. Conversely, the extension spring(s) **75** also may aid in the lowering of the height-of-cut by dampening the upward movement of the wheels, which, without the extension spring(s) **75**, would otherwise be abrupt due to the overall weight of the frame **72** and the various components coupled to frame **72** (e.g., the engine, hydrostatic transmission, etc.).

(46) Next, referring to FIGS. 9-14, a control assembly **100** for use with a walk-behind lawnmower (such as, e.g., lawnmower **10** described above) according to another aspect of the disclosure is shown. Control assembly **100** is configured to enable the user to control the speed and direction of the drive wheels of the lawnmower while maintaining at least one hand on or near the control handle at all times. More specifically, control assembly **100** includes a crossbar handle **101** positioned between a pair of extended arms **102a**, **102b**, which are each coupled to a frame (not shown) of the lawnmower. As shown in FIGS. 9-10, the extended arm **102a** may include a throttle control bracket **103**, which may be configured to support a throttle control lever (not shown) coupled to the engine. Additionally, an operator presence control bar **104** may be pivotally coupled to each of extended arms **102a**, **102b**, with an operator presence cable bracket **105** being provided on extended arm **102b** so as to support an operator presence cable **109** (shown in FIG. 10). The purpose and function of both the throttle control system and operator presence system utilizing operator presence control bar **104** are well-known and, thus, will not be described in further detail herein.

(47) Control assembly **100** also includes a control housing **106**, wherein control housing **106** is configured to support and/or protect various components for speed/direction control of the walk-behind lawnmower. For example, a speed control lever **107** may extend from a top surface of control housing **106** so as to be easily accessible by the user, with speed control lever **107** being pivotable about a pivot axis **116** relative to control housing **106** in order to allow the user to manually select a particular forward speed based on, e.g., terrain, mowing conditions, user comfort, etc. Additionally, a forward speed control handle **108** may be coupled to one or more primary arcuate brackets **120**, with arcuate bracket(s) **120** being coupled between respective walls **123**, **124** of the control housing **106** about a pivot axis **121**. One or more reverse speed control handles **110** may also be provided, with the one or more reverse speed control handles **110** being coupled to one or more respective secondary arcuate brackets **111** about a pivot axis **112**. As will be set forth in further detail below, the maximum forward speed of the lawnmower may be selected based upon the user-selected position of speed control lever **107**. Furthermore, the arcuate bracket(s) **120** and

the arcuate bracket(s) **111** may be linked together so as to provide both forward and reverse control inputs to the hydrostatic transmission (not shown) of the lawnmower via a single control linkage **132**.

(48) As illustrated in FIGS. **10-13**, speed control lever **107** extends from an asymmetric body portion **115**, with pivot axis **116** extending through asymmetric body portion **115**. A curved slot **117** may be formed within asymmetric body portion **115**. A post **118** may extend from a wall **124** of control housing **106** so as to at least partially pass through the curved slot **117**, with the post **118** being configured to form a stop at each respective end of the curved slot **117**, delineating the “minimum” and “maximum” speed control settings. Referring to FIG. **13**, a portion of the control housing **106** proximate the speed control lever **107** may include a forward speed range indicator **126**, providing the user with a visual indication of any number of possible forward speed settings (e.g., “maximum”, “3”, “2”, “minimum”, etc.). Furthermore, as shown in FIGS. **10, 12, and 13**, a compression spring assembly **127** may be provided so as to securely hold the speed control lever **107** (and asymmetric body portion **115**) in a desired position based on a user-selected forward speed setting, while still allowing for manipulation of the speed control lever **107** when enough torque is applied to overcome the compressive force of the spring.

(49) Referring FIGS. **11 and 13**, asymmetric body portion **115** includes a plurality of tooth-like projections **128a, 128b, 128c** which extend from a radial edge of the asymmetric body portion **115**. While three projections **128a, 128b, 128c** are shown in FIGS. **11 and 13**, it is to be understood that more or fewer projections may be utilized. Due to their position on the radial edge of asymmetric body portion **115**, projections **128a, 128b, 128c** (and/or the valleys therebetween) are configured to form a plurality of stop positions to limit the travel of forward speed control handle **108** and, more particularly, the travel of the one or more primary arcuate brackets **120** coupled to the forward speed control handle **108**.

(50) Specifically, the one or more primary arcuate brackets **120** may include a post **130** extending inwardly therefrom, wherein post **130** is sized and configured to engage the asymmetric body portion **115** at or between the projections **128a, 128b, 128c**, with the exact engagement point being dependent upon the user-defined speed setting of speed control lever **107**. The one or more primary arcuate brackets **120** are configured to pivot about the pivot axis **121**, thereby allowing post **130** extending therefrom to travel along an arcuate path as the user presses forward and downward upon forward speed control handle **108** to facilitate forward motion of the lawnmower. In FIGS. **10, 11, and 13**, speed control lever **107** is shown in its “maximum” speed position, with post **118** engaging with the end of slot **117** to prevent the asymmetric body portion **115** from further “forward” rotation. Accordingly, in this “maximum” speed position, the one or more primary arcuate brackets **120** and forward speed control handle **108** are allowed to move along a maximum-length arcuate path, thereby linearly displacing the control linkage **132** coupled to the one or more primary arcuate brackets **120** to a defined “maximum” extent, which, in turn, drives the hydrostatic transmission (not shown) at a defined “maximum” speed.

(51) In this “maximum” speed position, the post **130** will eventually engage with the asymmetric body portion **115** at or near, e.g., projection **128c**. However, as the user reduces the forward speed position by pulling back on speed control lever **107**, the shape of asymmetric body portion **115** is configured such that the arcuate path achievable by the one or more primary arcuate brackets **120** is limited. For example, if the user were to pull speed control lever **107** to its “minimum” setting, one or more of the projections **128a, 128b, 128c** are moved closer to the one or more arcuate brackets (and associated post **130**) due to the shape of asymmetric body portion **115**. Thus, in the “minimum” speed setting, as the user presses forward and downward upon forward speed control handle **108** to cause forward motion of the lawnmower, the post **130** contacts the asymmetric body portion **115** at or near one or more of the projections **128a, 128b, 128c** earlier along the arcuate path of the one or more primary arcuate brackets **120**, thereby reducing the linear displacement of control linkage **132**. Accordingly, as the travel of the control linkage **132** is limited, the input to the

hydrostatic transmission is similarly limited, thereby providing for a reduced forward speed.

(52) In the configurations shown in 9-14, the forward speed control handle **108** is shown as being positioned substantially near the crossbar handle **101**, slightly behind and below the crossbar handle **101** relative to the user. Due to this position, along with position of pivot axis **121** and the arcuate path of travel of the one or more primary arcuate brackets **120**, the forward speed control handle **108** allows for the user to use the palm and/or thumb of a single hand to control forward speed of the lawnmower, while still enabling the user to hold the crossbar handle **101** with the fingers of the same hand. Referring to FIG. **12**, the control handle **108** is shown as being located solely on the operator's right-hand side relative to the direction of forward travel. However, it is to be understood that the mounting bracket **122** which couples control handle **108** to the one or more primary arcuate brackets **120** may allow for reversible mounting such that control handle **108** may instead be located on the operator's left-hand side. Additionally and/or alternatively, while only a single-sided control handle **108** is shown, it is to be understood that the control handle **108** may be configured so as to have both a right-hand side and a left-hand side handle portion.

(53) Still referring to FIGS. **9-14**, as noted above, control assembly **100** may further include one or more reverse speed control handles **110**. As opposed to requiring the user to effectively “pull” backward on the control handle **108** to cause the hydrostatic transmission to drive the lawnmower in reverse, the one or more reverse speed control handles **110** provide for a more ergonomically pleasing mechanism to linearly displace the control linkage **132** in a reverse direction, thereby driving the lawnmower in reverse.

(54) As is best shown in FIGS. **10-12** and **14**, the one or more reverse speed control handles **110** may be coupled to one or more respective secondary arcuate brackets **111** about a pivot axis **112**. The secondary arcuate bracket(s) **111** may be pivotally coupled to the one or more primary arcuate brackets **120** by one or more links **113**. Thus, any movement of the secondary arcuate bracket(s) **111** results in corresponding movement of the arcuate bracket(s) **120**, and vice versa, which also moves control linkage **132**. Thus, the combination of, and interrelation between, the secondary arcuate bracket(s) **111** and the primary arcuate bracket(s) **120** provides both forward and reverse control inputs to the hydrostatic transmission (not shown) of the lawnmower via the single control linkage **132**.

(55) As shown in FIGS. **9-14**, the one or more reverse speed control handles **110** are positioned slightly in front of and slightly below the crossbar handle **101** relative to the operator. In this position, the operator may easily manipulate the reverse speed control handle(s) **110** using one or more fingers of one or more hands, while also maintaining a grip on crossbar handle **101**. This positioning of reverse speed control handle(s) **110** allows the operator to naturally “pull” the lawnmower in the reverse direction, without the need to release their grip on crossbar handle **101** and/or manipulate the forward speed control handle **108**.

(56) While FIGS. **9-12** illustrate a pair of reverse speed control handles **110**, it is to be understood that only a single reverse speed control handle **110** may be utilized, as is illustrated in FIGS. **13-14**.

(57) Next, referring to FIG. **15**, a hydrostatic transmission linkage assembly **140** in accordance with another aspect of the disclosure is shown. Hydrostatic transmission linkage assembly **140** may be configured for use with control assembly **100** described above so as to control the forward and/or reverse directional inputs to the hydrostatic transmission (not shown) from the control linkage **132**.

(58) Hydrostatic transmission linkage assembly **140** may include a pivoting member **141**, wherein pivoting member **141** includes an axle portion **146** pivotable between a first wall **144** and a second wall **145**, with each wall **144**, **145** extending from an upper surface of a lawnmower frame **168** substantially above the hydrostatic transmission (not shown) driving the rear wheels of the lawnmower. The pivoting member **141** includes a plurality of arms extending therefrom, with each arm configured to translate a control arm of the hydrostatic transmission (not shown) based on the directional input from the control linkage **132** and/or limit the rotational movement of the pivoting member **141**. Specifically, pivoting member **141** includes a first arm **148** extending therefrom,

which is configured to couple the control linkage **132** to the hydrostatic transmission about a pivot axis **150**. A second arm **142** and a third arm **154** extend from opposite lateral sides of axle portion **146**, with second arm **142** being coupled to a transmission linkage **152** at a pivot axis **153** such that rotation of pivoting member **141** about axle portion **146** leads to vertical translation of the transmission linkage **152**, thereby controlling the hydrostatic transmission (not shown) to drive in forward and/or reverse directions.

(59) Referring still to FIG. **15**, the first wall **144** further includes a pair of limit screws **155**, **156**. The length of limit screws **155**, **156** may be adjustable by any or all of the manufacturer, a fleet manager, a user, etc., so as to limit the rotational movement of the pivoting member **141** in multiple directions, thereby providing for additional maximum speed controls in both the forward and reverse directions. For example, as shown in FIG. **15**, limit screw **155** is shown as extending further downward than limit screw **156**, with limit screw **155** configured to control the maximum speed in the reverse direction and limit screw **156** configured to control the maximum speed in the forward direction. If control linkage **132** is “pulled” in the reverse direction by the user so as to cause reverse drive, the second arm **142** may only pivot upwards until the point where a top surface of the second arm **142** contacts the limit screw **155**, thereby limiting the speed of reverse travel. Conversely, if control linkage **132** is “pushed” in the forward direction by the user so as to cause forward drive, an upper surface of the third arm **154** may contact the limit screw **156**, thereby limiting the speed of forward travel. As noted above, the length/depth of limit screws **155**, **156** may be adjusted by the manufacturer, a fleet manager, and/or the user, dependent upon various factors.

(60) FIG. **15** also shows a fourth arm **158** coupled to axle portion **146** of pivoting member **141**. Fourth arm **158** is pivotally coupled, via a pivot axis **161**, to a spring biasing member **160**, wherein spring biasing member **160** is configured to bias the position of transmission linkage **152** to a “neutral” position if and when control linkage **132** is not being actively moved in the forward or reverse direction by the user. Thus, when the user releases both the forward speed control handle **108** and reverse speed control handle(s) **110**, as described above, the spring biasing member **160** returns the transmission linkage **152** to a neutral position such that the lawnmower is moved in neither the forward or reverse directions when not under active control by the user.

(61) Next, referring to FIG. **16**, the position of pivot axis **150** on first arm **148** is illustrated in relation to respective pivot joints **164**, **165** of extended arms **102b**, **102a**, respectively. As is shown, pivot axis **150** is substantially aligned along an axis C with pivot joints **164**, **165**. With this configuration, if the user were to pivot the extended arms **102a**, **102b** for transport, storage, etc., the distal end of control linkage **132** pivots about the same axis (i.e., axis C), thereby enabling the user control interface (extended arms and control assembly) to be pivoted without necessitating the decoupling of control linkage **132** from either the pivoting member **141** or the arcuate bracket(s) **120**. That is, if pivot axis **150** were not aligned with pivot joints **164**, **165**, the control linkage **132** would bind at a certain angular position, thereby limiting the folding position of extended arms **102a**, **102b**. However, with the pivot axis **150** aligned with pivot joints **164**, **165**, no such binding occurs.

(62) While FIGS. **1-16** illustrate various assemblies associated with a walk-behind lawnmower, it is to be understood that the features described herein may also be applicable to other types of outdoor power equipment, such as, e.g., aerators, seeders, turf cutters, brushcutters, rototillers, powered vacuums, powered leaf blowers, etc.

(63) The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The terminology used herein was chosen to best explain the principles of the embodiment, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. Outdoor power equipment comprising: a frame; a power head coupled to the frame; a transmission coupled to the frame and operatively coupled to the power head, wherein the transmission comprises a drive rotatable member configured to rotate about a first axis; a pair of rear drive wheels coupled at respective ends of a rear drive axle, wherein the rear drive axle is coupled to the drive rotatable member so that the rear drive axle, and the pair of rear drive wheels coupled thereto, rotate about a second axis different than the first axis; and a wheel height adjustment mechanism including: a first bell crank pivotally coupled to the frame proximate to a first rear wheel of the pair of rear drive wheels, wherein a first end of the first bell crank is coupled to the rear drive axle; a second bell crank pivotally coupled to the frame proximate to a second rear wheel of the pair of rear drive wheels; a third bell crank pivotally coupled to the frame, or to a mower deck fixedly coupled to the frame, at a location proximate a first front wheel; a fourth bell crank pivotally coupled to the frame, or to the mower deck fixedly coupled to the frame, at a location proximate a second front wheel; a first linkage extending between the first bell crank and the third bell crank; a second linkage extending between the second bell crank and the fourth bell crank; and a first front pivoting linkage and a second front pivoting linkage pivotally coupled to the first front wheel and the second front wheel at a position vertically in-line with a pivoting location of the third bell crank and the fourth bell crank, respectively, so as to form a 4-bar linkage-type arrangement.
2. The outdoor power equipment of claim 1, wherein a second end of the first bell crank is coupled to a retractable pin member.
3. The outdoor power equipment of claim 2, wherein a portion of the frame includes a plurality of incrementally-spaced through-holes, and wherein the retractable pin member is configured to selectively engage with a selected one of the plurality of incrementally-spaced through-holes.
4. The outdoor power equipment of claim 1, further comprising at least one extension spring, wherein a first end of the at least one extension spring is coupled to the frame or the mower deck and a second end of the at least one extension spring is coupled to the first linkage or the second linkage.
5. The outdoor power equipment of claim 1, wherein the first front wheel is a first caster wheel assembly and the second front wheel is a second caster wheel assembly.
6. The outdoor power equipment of claim 1, wherein the transmission is coupled to the power head via a drive belt.
7. The outdoor power equipment of claim 6, further comprising the mower deck positioned between the pair of rear drive wheels and both the first front wheel and the second front wheel.
8. The outdoor power equipment of claim 7, wherein the transmission, the power head, and the mower deck are fixed relative to the frame.
9. The outdoor power equipment of claim 8, wherein an angle of the drive belt remain constant regardless of a height of the pair of rear drive wheels and the pair of front wheels relative to the frame.
10. A walk-behind lawnmower comprising: a frame; a power head coupled to the frame; a mower deck coupled to the frame; a transmission coupled to the frame and operatively coupled to the power head, wherein the transmission comprises a drive member configured to rotate about a first axis; a front wheel; a pair of rear drive wheels coupled at respective ends of a rear drive axle, wherein the rear drive axle is coupled to the drive member so that the rear drive axle, and the pair of rear drive wheels coupled thereto, rotate about a second axis; and a wheel height adjustment mechanism including: at least one rear bell crank pivotally coupled to the frame, wherein a first end of the at least one rear bell crank is coupled to the rear drive axle such that movement of the rear drive axle relative to the frame does not affect tension of a drive between the drive member and the

rear drive axle; at least one linkage extending between the at least one rear bell crank and a front wheel bell crank coupled to a front wheel such that height of the front wheel relative to the frame is adjusted simultaneously with any height adjustment of the pair of rear drive wheels; and a front pivoting linkage pivotally coupled between the frame and a wheel bracket of the front wheel, wherein the front wheel bell crank and the front pivoting linkage form a four-bar linkage between the front wheel and the frame.

11. The walk-behind lawnmower of claim 10, further comprising a retractable pin member coupled to a second end of the at least one rear bell crank, wherein the retractable pin member is configured to selectively engage with a selected one of a plurality of through-holes formed in a portion of the frame.

12. The walk-behind lawnmower of claim 10, further comprising at least one extension spring, wherein a first end of the at least one extension spring is coupled to the frame or the mower deck and a second end of the at least one extension spring is coupled to the at least one linkage.

13. The walk-behind lawnmower of claim 10, wherein the at least one rear bell crank is pivotally coupled to the frame at a location proximate a first rear wheel of the pair of rear drive wheels, and a second bell crank is pivotally coupled to the frame at a location proximate a second rear wheel of the pair of rear drive wheels.
